A Hydrodynamic Approach to Macroscopic Gravity: Thermodynamic Equilibrium via Viscous Dissipation and Unit-Quaternion Tensors

Jung Soo Kim
Arkhenomia Institute
ORCID: 0009-0006-5140-0604

Archival Note: This foundational monograph is consolidated into Part 1 of the 2026 *Grand Unified Framework* series (*A Grand Unified Framework for Macroscopic Gravity-Fluid Dynamics and 0-Point Equilibrium Operation*).

Abstract

This monograph presents a continuum-mechanical and hydrodynamic framework for macroscopic gravito-fluid equilibrium, bridging relativistic stress-energy tensor geometry and three-dimensional Navier–Stokes fluid dynamics. By modeling macroscopic rotational and gravitational stress fields within a bounded planetary control volume $\Omega \subset \mathbb{R}^3$, we introduce a dimensionally calibrated unit-quaternion rotational stress tensor ($q_c \in \mathrm{Sp}(1) \cong SU(2)$) that eliminates the polar coordinate singularities ($\theta \in \{0, \pi\}$, gimbal lock) inherent to Euler-angle formulations. Furthermore, to prevent non-physical gravitational implosion in idealized point-mass density models, an isotropic hydrostatic pressure limit tensor ($-P\boldsymbol{I}$) is coupled with a radial density stratification profile ($\rho(r)$). Through the Virial tensor divergence identity, the macroscopic boundary stress integral over $\partial\Omega$ is rigorously mapped to an interior visco-elastic and hydrodynamic dissipation balance, where kinematic viscous dissipation ($-\nu\Delta\boldsymbol{u}$) counteracts non-linear convective vortex acceleration ($\boldsymbol{u} \times (\nabla \times \boldsymbol{u})$). This formulation establishes a singularity-free, dimensionally homogeneous (Pa) macroscopic stress closure for rotating continuum systems.

Contents

1 Introduction **2**

2 Theoretical Framework and Hydrodynamic Stress Equations **2**
2.1 Step 1: Classical Point-Mass Interaction (Newtonian Baseline) 2
2.2 Step 2: Boundary Stress Contraction (Geometric Continuum Basis) 2
2.3 Step 3: Unit-Quaternion Rotational Stress and Hydrostatic Equilibrium 3
2.4 Step 4: Viscous Dissipation and Virial Thermodynamic Equilibrium 3

3 Analytical Stability and Singularity Mitigation Mechanisms **3**
3.1 Elimination of Polar Coordinate Singularities via $\mathrm{Sp}(1)$ Quaternions 3
3.2 Hydrostatic Prevention of Point-Mass Implosion 4
3.3 Dissipative Attenuation of Turbulent Kinetic Energy (TKE) 4

4 Multi-Scale Topological Extension to Periodic Continuum Domains **4**

5 Conclusion **4**

Acknowledgments and Declarations **4**

© 2026 Jung Soo Kim (Arkhenomia Institute). Licensed under CC BY-NC 4.0.

1 Introduction

The interplay between continuum mechanics, gravitational potential theory, and fluid dynamics has historically provided the foundation for modeling rotating celestial bodies and geophysical flows [1, 2]. While macroscopic gravitational fields are classically described through geometric curvature tensors or Newtonian potential gradients, direct numerical and analytical modeling of three-dimensional rotating planetary interiors and envelopes exposes two persistent mathematical obstructions: (i) coordinate singularities at the rotational poles ($\theta \in \{0, \pi\}$) in spherical Euler-angle parameterizations, and (ii) unbounded density and stress concentration in unregularized point-mass or inviscid vortex models.

To address these limitations within a self-contained continuum framework, this study formulates macroscopic gravito-fluid equilibrium in terms of a volume-normalized Virial Cauchy stress functional. Rather than treating gravitational and rotational forces purely as external point-source potentials, our approach incorporates the internal hydrostatic resistance, rotational angular momentum flux, and viscous dissipation capacity of the continuum medium directly into a unified rank-2 stress tensor.

Specifically, by mapping the axial vorticity vector $\boldsymbol{\omega} = \nabla \times \boldsymbol{u}$ into the Lie algebra $\mathfrak{sp}(1)$ of pure quaternions and applying the unit-quaternion sandwich action $\tilde{\boldsymbol{\omega}} = q_c \boldsymbol{\omega} q_c^*$ ($q_c \in \mathrm{Sp}(1)$) [3], we construct a symmetric, positive semi-definite rotational stress tensor that is smooth across the entire celestial sphere. Coupling this rotational stress with a radial density profile $\rho(r)$, an isotropic hydrostatic pressure tensor $-P\boldsymbol{I}$, and interior Navier–Stokes viscous dissipation [2] yields an analytically tractable thermodynamic equilibrium closure that prevents finite-time energy divergence.

2 Theoretical Framework and Hydrodynamic Stress Equations

We construct the macroscopic stress closure in four progressive stages, advancing from classical point-mass mechanics to a boundary-to-interior Virial tensor equilibrium over a bounded control volume $\Omega \subset \mathbb{R}^3$ with volume $V_\Omega = |\Omega| > 0$ and Lipschitz boundary $\partial\Omega$ with outward unit normal $\hat{\boldsymbol{n}}$.

2.1 Step 1: Classical Point-Mass Interaction (Newtonian Baseline)

In classical mechanics, gravitational attraction between point masses m_1, m_2 separated by distance $r > 0$ is governed by the inverse-square law:

$$\boldsymbol{F} = -G\frac{m_1 m_2}{r^2}\hat{\boldsymbol{r}}. \tag{1}$$

While effective for exterior Keplerian orbital mechanics, Equation (1) diverges as $r \to 0$ and does not capture internal continuum shear stresses, rotational deformation, or viscous energy dissipation within a finite planetary body.

2.2 Step 2: Boundary Stress Contraction (Geometric Continuum Basis)

In continuum and relativistic field theories [1], local conservation of momentum is expressed through a rank-2 stress-energy or Cauchy stress tensor. Integrating the normal traction of an effective influx stress tensor $\boldsymbol{R}_\chi$ against a spatial moment arm $\boldsymbol{M}$ over the control boundary $\partial\Omega$ defines the baseline geometric moment functional:

$$\boldsymbol{\Phi}_{\mathrm{geo}} = \frac{1}{V_\Omega}\oint_{\partial\Omega} \boldsymbol{M} \otimes (\boldsymbol{R}_\chi \cdot \hat{\boldsymbol{n}})\, \mathrm{d}\Sigma. \tag{2}$$

© 2026 Jung Soo Kim (Arkhenomia Institute). Licensed under CC BY-NC 4.0.

However, when parameterized in standard spherical coordinates without rotational regularization or hydrostatic counter-pressure, Equation (2) remains vulnerable to polar coordinate degeneracies ($\sin\theta \to 0$) and unbuffered compressive collapse.

2.3 Step 3: Unit-Quaternion Rotational Stress and Hydrostatic Equilibrium

To eliminate the polar gimbal-lock singularity and arrest compressive collapse, we introduce the angular momentum calibration factor $\mathcal{K}_\omega > 0$ ($[\mathrm{m}^2]$) and the unit quaternion $q_\mathrm{c} = q_0 + q_1\boldsymbol{i} + q_2\boldsymbol{j} + q_3\boldsymbol{k} \in \mathrm{Sp}(1) \cong SU(2)$ satisfying $\|q_\mathrm{c}\|^2 = 1$ and $q_\mathrm{c}^* = q_0 - q_1\boldsymbol{i} - q_2\boldsymbol{j} - q_3\boldsymbol{k}$ [3]. Identifying the vorticity vector $\boldsymbol{\omega} = \nabla \times \boldsymbol{u} \in \mathbb{R}^3$ with a pure quaternion, the rotated vorticity $\tilde{\boldsymbol{\omega}} = q_\mathrm{c}\boldsymbol{\omega}q_\mathrm{c}^*$ preserves the Euclidean norm $|\tilde{\boldsymbol{\omega}}|^2 = |\boldsymbol{\omega}|^2$. Taking its dyadic square and coupling with the radial density stratification $\rho(r)$ ($[\mathrm{kg}\cdot\mathrm{m}^{-3}]$) and isotropic hydrostatic pressure $-P\boldsymbol{I}$ ($[\mathrm{Pa}]$) yields the dimensionally homogeneous intermediate Virial flux tensor:

$$\boldsymbol{\Phi}_\mathrm{hydro} = \frac{1}{V_\Omega}\oint_{\partial\Omega} \boldsymbol{r} \otimes \left(\left[\mathcal{K}_\omega \rho(r) \left(q_\mathrm{c}\boldsymbol{\omega}q_\mathrm{c}^*\right)^{\otimes 2} + \boldsymbol{R}_\mathrm{X} - P\boldsymbol{I} \right] \cdot \hat{\boldsymbol{n}} \right) \mathrm{d}\Sigma. \tag{3}$$

Here, every bracketed constituent carries exact SI stress dimensions ($[\mathrm{kg}\cdot\mathrm{m}^{-1}\cdot\mathrm{s}^{-2} = \mathrm{Pa}]$). The symmetric dyadic tensor $\boldsymbol{\sigma}_\mathrm{rot} = \mathcal{K}_\omega\rho(r)(q_\mathrm{c}\boldsymbol{\omega}q_\mathrm{c}^*)^{\otimes 2}$ is positive semi-definite with trace $\mathrm{tr}(\boldsymbol{\sigma}_\mathrm{rot}) = \mathcal{K}_\omega\rho(r)|\boldsymbol{\omega}|^2$, providing centrifugal and hydrostatic structural support across Ω.

2.4 Step 4: Viscous Dissipation and Virial Thermodynamic Equilibrium

While Equation (3) establishes static morphological stability, continuous rotational forcing injects turbulent kinetic energy (TKE) into the fluid envelope and interior. To prevent exponential energy divergence and close the macroscopic energy budget, we incorporate the rank-4 Kakeya directional projection $\mathbb{P}_\mathrm{Kakeya}$ [5], the dimensionless linkage factor Γ_0, and the deviatoric efflux/viscous stress tensor $\boldsymbol{T}_\mathrm{X}$. Applying the Gauss divergence theorem componentwise ($\frac{1}{V_\Omega}\oint_{\partial\Omega} x_i\sigma_{jk}n_k\,\mathrm{d}\Sigma = \frac{1}{V_\Omega}\int_\Omega(\sigma_{ji} + x_i\partial_k\sigma_{jk})\,\mathrm{d}V$) maps the boundary traction integral to the interior visco-elastic and hydrodynamic relaxation balance:

$$\begin{aligned}\boldsymbol{\Phi}_\Omega(t) &= \frac{1}{V_\Omega}\,\mathrm{Re}\left\{ \oint_{\partial\Omega} \boldsymbol{r} \otimes \left(\left(\mathbb{P}_\mathrm{Kakeya} : \left[\mathcal{K}_\omega\rho(r)\left(q_\mathrm{c}\boldsymbol{\omega}q_\mathrm{c}^*\right)^{\otimes 2} + \Gamma_0\boldsymbol{R}_\mathrm{X} - P\boldsymbol{I} + \boldsymbol{T}_\mathrm{X} \right] \right) \cdot \hat{\boldsymbol{n}} \right) \mathrm{d}\Sigma \right\} \\ &= \boldsymbol{\Phi}_\mathrm{mantle} - \frac{1}{V_\Omega}\int_0^t \int_\Omega \boldsymbol{\mathcal{R}}_\mathrm{diss} \cdot \left(\boldsymbol{u} \times (\nabla \times \boldsymbol{u}) - \nu\Delta\boldsymbol{u}\right) e^{-\lambda(t-\tau)}\,\mathrm{d}V\,\mathrm{d}\tau. \end{aligned} \tag{4}$$

In Equation (4), $\boldsymbol{u}(\boldsymbol{x},\tau)$ is the continuum velocity field, $\boldsymbol{u} \times (\nabla \times \boldsymbol{u}) = \boldsymbol{u} \times \boldsymbol{\omega}$ is the Lamb vortex acceleration vector, $\nu > 0$ is the kinematic viscosity, $\lambda > 0$ is the visco-elastic relaxation rate, $\boldsymbol{\Phi}_\mathrm{mantle}$ is the baseline equilibrium stress buffer, and $\boldsymbol{\mathcal{R}}_\mathrm{diss}$ is the rank-3 moment-mapping tensor converting interior momentum dissipation density into macroscopic stress units (Pa).

3 Analytical Stability and Singularity Mitigation Mechanisms

We now summarize the three structural stability properties guaranteed by the governing formulation (4) in computational fluid dynamics (CFD) and geophysical continuum applications.

3.1 Elimination of Polar Coordinate Singularities via $\mathrm{Sp}(1)$ Quaternions

In standard 3D spherical-grid discretizations, Euler-angle rotation matrices introduce $\csc\theta$ metric factors that diverge at the poles ($\theta = 0, \pi$). Because the unit-quaternion group $\mathrm{Sp}(1) \cong S^3$ forms a smooth double cover of $SO(3)$ without local chart singularities [3], discretizing the rotational stress via $(q_\mathrm{c}\boldsymbol{\omega}q_\mathrm{c}^*)^{\otimes 2}$ ensures bounded, C^∞-smooth tensor components across all latitudes.

© 2026 Jung Soo Kim (Arkhenomia Institute). Licensed under CC BY-NC 4.0.

3.2 Hydrostatic Prevention of Point-Mass Implosion

Unlike unregularized self-gravitating dust models where $\rho \to \infty$ in finite time, the coupling between the stratified density profile $\rho(r)$ and the isotropic pressure buffer $-P\boldsymbol{I}$ in Equation (3) enforces a positive normal stress barrier against radial compression, preserving control-volume regularity.

3.3 Dissipative Attenuation of Turbulent Kinetic Energy (TKE)

In vorticity-dominated regimes ($\boldsymbol{\omega} = \nabla \times \boldsymbol{u}$), non-linear vortex stretching and Lamb acceleration $\boldsymbol{u} \times \boldsymbol{\omega}$ drive energy transfer toward small spatial scales. Within the volumetric memory integral of Equation (4), the molecular Laplacian dissipation $-\nu\Delta\boldsymbol{u}$ and exponential relaxation kernel $e^{-\lambda(t-\tau)}$ damp high-wavenumber fluctuations, driving the macroscopic Virial stress $\boldsymbol{\Phi}_\Omega(t)$ asymptotically toward the equilibrium buffer $\boldsymbol{\Phi}_{\text{mantle}}$ [4].

4 Multi-Scale Topological Extension to Periodic Continuum Domains

While Equation (4) establishes macroscopic thermodynamic equilibrium via linear viscosity $-\nu\Delta\boldsymbol{u}$, extreme high-Reynolds-number geophysical flows require adaptive non-linear strain-rate closures and microscopic vorticity regularization on boundaryless periodic domains ($\mathbb{T}^3$). By embedding the macroscopic Virial stress balance into a multi-scale topological framework—combining adaptive L^q strain-rate drag tensors in the macroscopic continuum with localized zero-point counter-strain projections in the microscopic core—the system achieves unconditional enstrophy regularity across scales.

5 Conclusion

This monograph has formulated a singularity-free hydrodynamic stress closure for macroscopic gravito-fluid systems. By combining unit-quaternion rotational kinematics ($q_c \in \mathrm{Sp}(1)$), hydrostatic pressure stabilization ($-P\boldsymbol{I}$), and Virial boundary-to-interior viscous dissipation, the model resolves polar coordinate degeneracies and compressive collapse within a dimensionally homogeneous (Pa) tensor equation. These foundational mechanisms form the macroscopic pillar of Part 1 of the 2026 *Grand Unified Framework* series.

Acknowledgments

The author expresses sincere gratitude to Researcher He Ra Shin for her insightful discussions and steadfast support throughout the course of this work. Her foundational inspiration contributed significantly to the theoretical formulation and formal realization of this research.

Declarations

Data and Code Availability. All mechanized Lean 4 formal verification suites, symbolic CAS scripts, numerical solvers, and benchmark datasets supporting the findings of this study are publicly archived via the canonical repository (`https://github.com/arkhenomia/arkheion`) and the dataset archive (`https://huggingface.co/datasets/arkhenomia/arkheion`).

Licensing and Governance. This manuscript and all associated formal proof scripts, source codes, numerical solvers, and datasets are licensed under the **Creative Commons Attribution-NonCommercial 4.0 International License (**`CC BY-NC 4.0`**)** (`https://creativecommons.org/`

licenses/by-nc/4.0/). Non-commercial academic research, educational use, and independent verification are freely permitted with proper attribution and primary reference citation. All disclosed mathematical equations and tensor architectures constitute published Prior Art against third-party appropriation (with all domestic and international patent rights expressly reserved by the original author, Arkhenomia Institute, and Arkhenetics), and all licensing conditions, commercial usage definitions, and governance interpretations are governed by the latest version of LICENSE.txt jointly established by the original author (Jung Soo Kim, ORCID: 0009-0006-5140-0604), Arkhenomia Institute, and Arkhenetics, and officially published via the canonical repository (https://github.com/arkhenomia/arkheion) and the official websites (https://arkhenomia.org, https://arkhenetics.com), which supersedes and prevails over any prior notices or earlier distribution drafts.

References

[1] A. Einstein, "Die Grundlage der allgemeinen Relativitätstheorie," *Annalen der Physik*, vol. 354, no. 7, pp. 769–822, 1916.

[2] G. G. Stokes, "On the theories of the internal friction of fluids in motion, and of the equilibrium and motion of elastic solids," *Transactions of the Cambridge Philosophical Society*, vol. 8, pp. 287–319, 1845.

[3] J. B. Kuipers, *Quaternions and Rotation Sequences: A Primer with Applications to Orbits, Aerospace, and Virtual Reality*. Princeton, NJ: Princeton University Press, 1999.

[4] T. Padmanabhan, "Classical and quantum thermodynamics of horizons in spherically symmetric spacetimes," *Classical and Quantum Gravity*, vol. 19, no. 21, pp. 5387–5408, 2002.

[5] S. Kakeya, "Some problems on maximum and minimum regarding ovals," *Tohoku Mathematical Journal*, vol. 11, pp. 92–100, 1917.

© 2026 Jung Soo Kim (Arkhenomia Institute). Licensed under CC BY-NC 4.0.